Dr. Thejus Mahajan

Contact Information

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Career Interests

To apply expertise in computational modeling, data science, and bioinformatics to address complex challenges in clinical and biological research. Skilled in multivariate statistical modeling, biostatistics, and developing analysis pipelines using R (Bioconductor, Tidymodels) and Python. Actively transitioning into clinical bioinformatics through specialized training on patient datasets and biostatistical methods for health research.

Work Experience

- **CQ Beratung + Bildung, Berlin, Germany**
  Further Training – Application-related Bioinformatics and Biostatistics
  August 2025 – Present

  - Shell (BASH), Python (object-oriented), and programming methodology.
  - Bioinformatics databases (NCBI, Biopython, SQL).
  - Sequence analysis, structural bioinformatics, and NGS analysis (Nextflow, Galaxy).
  - Applied biostatistics with real and simulated patient datasets (R/Bioconductor, ANOVA, PCA, HCA, survival analysis).
  - Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis — automated NCBI download, MAFFT alignment, and trimAl trimming with Conda-managed dependencies (GitHub).

- **HealthTwiSt GmbH, Berlin, Germany**
  Internship – Machine Learning in R
  February 2026 – April 2026

  - Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ˜300 clinics. Reduced to 1,348 lines (−26.5%) with byte-identical output verified via `identical()`.
  - Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
  - Replaced 357 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.
  - Created the `degirtools` R package (9 exported functions, roxygen2 docs, `devtools::check()` passing) by extracting reusable logic from a 767-line helper file.
  - Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review — demonstrating production-code discipline.
  - Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io with GDPR-safe synthetic data.
  - Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.

- **Helmholtz-Zentrum Hereon, Geesthacht, Germany**
  Guest Scientist – Computational Biologist & Systems Modeler
  May 2025 – October 2025

- Spearheaded research modeling evolutionary processes in marine environments.
- Simulated long-term impacts of environmental changes on ecosystems.
- Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

- **University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences**
  Post-doctoral Scientist – Marine Ecosystem Modelling
  August 2021 – January 2025

- Developed the "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework.
- Engineered models in Fortran and analyzed large NetCDF datasets using Python in Linux.
- Statistical analysis and visualization of ecological time-series data using R.

- **Université Paris-Saclay, Institut des Sciences Moléculaires d'Orsay, France**
  Ph.D. Researcher in Astrochemistry
  October 2015 – September 2018

- Led atomic collision experiments and processed terabytes of raw data.
- Utilized C++ for signal processing and algorithm development.
- Developed a molecular fragmentation model for the KIDA database.
- **Thesis**: "Excitation and fragmentation of C_nN^+ (n=1-3) molecules in collisions with He atoms at intermediate velocity; fundamental aspects and application to astrochemistry."
- **Advisor**: Dr. Karine Béroff

Education

- **Ph.D. in Astrochemistry** (2015 – 2018)
  Institut des Sciences Moléculaires (ISMO), Université Paris-Saclay, France

- **M.Sc. in Physics** (July 2012 – January 2015)
  National Institute of Technology Calicut, India
  CGPA: 8.2/10

Skills & Competencies

Programming:	Python (Pandas, NumPy, Scikit-learn), R (Bioconductor, Tidymodels, ggplot2, R Shiny, devtools/roxygen2), Fortran, SQL, BASH
Bioinformatics:	NGS Analysis, Galaxy, Sequence Alignment, Structural Bioinformatics, Nextflow
Biostatistics:	Multivariate Analysis, PCA, ANOVA, Dimensionality Reduction, Markov Chains
Data Analysis:	Large dataset processing (NetCDF, HDF5), Data visualization, Pipeline development
Tools:	Linux/Unix, Git, HPC clusters, LaTeX, Microsoft Office

Languages

English:	Proficient (speaking, reading, writing)
German:	B1 (Goethe Certified)
French:	Intermediate
Malayalam:	Native
Tamil, Hindi:	Proficient

Training & Courses

- HPC Training at Jülich Supercomputing Center (JSC)
- MITx: 7.00x Introduction to Biology – The Secret of Life

Publications

Google Scholar: `https://scholar.google.com/citations?hl=en&user=PJkZwAwAAAAJ`

1. T. Mahajan, et al. *Journal of Physics: Conference Series*, 1412:142026, 2020.
2. T. Idbarkach, et al. *Astronomy & Astrophysics*, 628:A75, 2019.
3. T. Mahajan, et al. *Journal of Physics B*, 2019.
4. T. Idbarkach, et al. *Journal of Physics B*, 51(24):245201, 2018.
5. T. Idbarkach, et al. *Journal of Physics: Conference Series*, 1412(11):112028, 2020.

References

- **Dr. Andreas Busjahn**
  HealthTwiSt GmbH, Berlin
  Email: `andreas@busjahn.net`
  Professional relation: Biostatistics instructor and internship supervisor
- **Prof. Dr. Kai Wirtz**
  Helmholtz-Zentrum Hereon, Ökosystemmodellierung
  Email: `kai.wirtz@hereon.de`
  Professional relation: Supervisor during Guest Scientist position
- **Prof. Dr. Elisa Schaum**
  University of Hamburg, Institute of Marine Ecosystem and Fisheries Sciences
  Email: `elisa.schaum@uni-hamburg.de`
  Professional relation: Collaborator during Post-doctoral research